MILITARY SPECIFICATION

CONNECTOR, ELECTRICAL

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of connector, electrical used on M454, M455 and T5123E1 projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	-Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1169	-Packaging, Packing and Marking for Shipments of Inert Ammunition Components

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DRAWINGS

PICATINNY ARSENAL

8808362-Packing and Marking for Storage and Shipment
8883535-Bag, Packing, for Repair Parts
9221103-Connector, Electrical
9220310-1-Screw, Connector
9220317-Pin, Guide, Connector
9220316-Connector Body Assembly
9220319-Body Connector
9220320-Insulator, Connector
9220323-Sleeve
9220322-Adhesive Sealant
ET9221103-Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The connector shall comply with all requirements specified on the drawing (dwg) 9221103, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Tool manufacturing control (T.M.C.). When T.M.C. is specified herein, five consecutive parts which comply with drawing dimensions will constitute proper initial tool control acceptance. During normal production, five parts shall be selected at the start and at the end of each days production (for a minimum of 10 parts) and examined as specified in the applicable paragraph in 4.3.2.1. In addition, five consecutive parts shall be examined as specified in the applicable paragraph in 4.3.2.1 whenever the manufacturing tool requires maintenance. Failure of any part to comply with any of the examinations specified in 4.3.2.1 shall reject the lot of production under consideration.

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3.4 Workmanship.

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable

4.3 Quality Conformance Inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produces by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I, II, III, and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major Characteristics for War Reserve Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	421-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276

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Table I (cont)
Major Characteristics for War Reserve Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics for War Reserve Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-78	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Major Characteristics for Training and Test Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-39	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table IV
Minor Characteristics for Training and Test Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Adhesive, sealant (see dwg. 9220322 covering a detail of dwg. 9220316).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for adhesive in accordance with dwg. 9220322.	Visual
Minor: None defined.		

4.3.2.1.2 Sleeve (see dwg. 9220323 covering a detail of dwg. 9220316).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Right outside diameter	Gage
102.	Inside diameter	Gage
103.	Left outside diameter	Gage
104.	Total length	Gage
105.	Length of right diameter	Gage
106.	Material (see 3.1)	Visual
Minor:		
201.	Surface finish	Visual
202.	Workmanship (see 3.4)	Visual/ Manual

4.3.2.1.3 Insulator, connector (see dwg. 9220320 covering a detail of dwg. 9220316).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Thickness	T.M.C. (see 3.3)
102.	Length	T.M.C. (see 3.3)
103.	Large radius	T.M.C. (see 3.3)
104.	Small radius	T.M.C. (see 3.3)

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Categories	Characteristics	Method of Inspection
105.	Diameter of 2 end holes	T.M.C. (see 3.3)
106.	Location of 2 end holes	T.M.C. (see 3.3)
107.	Diameter of center hole	T.M.C. (see 3.3)
108.	Location of center hole	T.M.C. (see 3.3)
109.	Diameter of counterbored holes	T.M.C. (see 3.3)
110.	Location of counterbored holes	T.M.C. (see 3.3)
111.	Diameter of counterbores	T.M.C. (see 3.3)
112.	Concentricity of counterbores w/holes	T.M.C. (see 3.3)
113.	Depth of counterbores	T.M.C. (see 3.3)
114.	Countersink present	Visual
115.	No tear, split or crack	Visual
116.	Material (see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.4 Body connector (see dwg. 9220319 covering a detail of dwg. 9220316).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Length	T.M.C. (see 3.3)
102.	Depth of counterbore center hole	T.M.C. (see 3.3)
103.	Diameter of counterbore center hole	T.M.C. (see 3.3)
104.	Diameter of center hole	T.M.C. (see 3.3)
105.	Thickness	T.M.C. (see 3.3)
106.	Large radius	T.M.C. (see 3.3)
107.	Small radius	T.M.C. (see 3.3)
108.	Pitch diameter of thread in 2 end hole	T.M.C. (see 3.3)
109.	Minor diameter of thread in 2 end hole, max.	T.M.C. (see 3.3)
110.	Location of 2 end holes	T.M.C. (see 3.3)
111.	Diameter of small holes	T.M.C. (see 3.3)
112.	Location of small holes	T.M.C. (see 3.3)
113.	Diameter of top counterbores on small holes	T.M.C. (see 3.3)
114.	Concentricity of top counterbores with small holes	T.M.C. (see 3.3)
115.	Depth of top counterbores	T.M.C. (see 3.3)
116.	Depth of bottom counterbores on small holes	T.M.C. (see 3.3)

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Categories	Characteristics	Method of Inspection
	117. Diameter of bottom counterbores with small holes	T.M.C. (see 3.3)
	118. Concentricity of bottom counterbores with small holes	T.M.C. (see 3.3)
	119. No tear, split or crack	Visual
	120. Flash removed	Visual/Manual
	121. Material (see 3.1)	Visual
Minor:		
	201. Workmanship (see 3.4)	Visual/Manual

4.3.2.1.5 Connector body assembly (see 9220316 covering a detail of dwg. 9221103).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
	101. All parts present	Visual
	102. Correct assembly	Visual
	103. Material (insulation tubing)(see 3.1)	Visual
Minor:		
	201. Marking, present, complete, correct, and identifiable	Visual
	202. Workmanship (see 3.4)	Visual

4.3.2.1.6 Pin, guide connector (see dwg. 9220317 covering a detail of dwg. 9221103).

Categories	Characteristics	Method of Inspection
Critical: None defined.		

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Categories	Characteristics	Method of Inspection
Major:		
	101. Left diameter	Gage
	102. Length	Gage
	103. Length of thread (incl. relief)	Gage
	104. Length of right diameter	Gage
	105. Right diameter	Gage
	106. Pitch diameter of thread	Gage
	107. Major diameter of thread, max	Gage
	108. Thread relief present	Visual
	109. Material (see 3.1)	Visual
Minor:		
	201. Radius present	Visual
	202. Surface finish	Visual
	203. Workmanship (see 3.4)	Visual/Manual

4.3.2.1.7 Screw, connector (see dwg. 9220310-1 covering a detail of dwg. 9221103).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
	101. Pitch diameter of thread	Gage
	102. Major diameter of thread, min	Gage
	103. Thread clearance, max	Gage
	104. Pilot diameter	Gage
	105. Distance from left face to left shoulder	Gage
	106. Total length	Gage
	107. Length of head	Gage
	108. Width of slot	Gage
	109. Depth of slot	Gage
	110. Pilot shoulder clearance present	Visual
	111. Diameter of head	Gage
	112. Material (see 3.1)	Visual

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Categories	Characteristics	Method of Inspection
Minor:		
	201. Location of screw slot	Gage
	202. Chamfer present	Visual
	203. Surface finish	Visual
	204. Workmanship (see 3.4)	Visual/Manual

4.3.2.1.8 Connector, electrical (see dwg. 9221103).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
	101. Distance from top of pins to bottom side of connector body assy	Gage
	102. All parts present and properly assembled	Visual
Minor:		
	201. Workmanship (see 3.4)	Visual/Manual

4.3.2.1.9 Bag, Packing, for Repair Parts (see dwg. 8883535 covering dwg. part number 8883535-2 and 5.2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
	101. Material (see 3.1)	Visual
	102. Length before packing, min	Measure*
	103. Width before packing, min	Measure*
	104. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
	105. Properly packed	Visual
	106. No tear, split, or crack in bag	Visual

*Measure with suitable scale.

Minor:		
	201. Marking	Visual
	202. Workmanship (see 3.4)	Visual/Manual

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4.3.2.1.10 Packing and Marking for Storage and Shipments (see dwg. 8808362 and 5.2).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Cross load weight, max.	Measure*
	102. Contents tight in package.	Manual
	103. Correctly packed.	Visual/Manual
	104. Package material (see 3.1).	Visual

*Determine weight with a suitable scale.

Minor:		
	201. Marking.	Visual
	202. Workmanship.(see 3.4).	Visual/Manual

4.3.2.1.11 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packaging, packing and marking in accordance with MIL-STD-1169.	Visual/Manual

Minor: None defined.

4.3.3 Testing. Not applicable.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET9221103. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2)

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. Not applicable.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The connector shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The connector shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The connector shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The connector shall be packaged in accordance with dwg. 8883535 and 9808362 using the specified moisture barrier.

5.2.1.2 Level C. The connector shall be packaged in accordance with dwg. 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The connector shall be packed in accordance with dwg. 8883535 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg. 8883535 and 8808362 as spplicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

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a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
d. Part used for War Reserve, Training or Test Projectile.

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.3.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A 029

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